

Chapter 1

Lambek linguistically: Towards a category theoretic semantics

Miriam S. Schmidt

Georg-August University Göttingen

This paper develops a category theoretic framework for natural language semantics grounded in the Curry-Howard-Lambek correspondence, with particular focus on intensional and hyperintensional phenomena. I argue that both classical simple type theories and modern record type theories inherit a fundamentally extensional notion of meaning from functional completeness, forcing intensional distinctions to be encoded through additional parameters, worlds, or type-theoretic complexity. In contrast, the present framework weakens functional completeness from uniqueness to representability within intensional domains, thereby allowing morphisms to differ structurally despite identical extensional behavior.

Building on a two-domain architecture separating extensional and intensional objects, intensionality and its effects can be analyzed locally. Propositions then are not identified as truth-valued functions but as structured categorical diagrams inducing semantic evaluations. This permits a strictly relational notion of meaning grounded in Yoneda embeddings rather than extensional identity alone. The resulting system exemplarily derives factivity and perspective-dependence as structural constraints on morphisms and intensional subcategories, rather than as lexical stipulations or contextual parameters. The paper thus proposes a genuinely category theoretic foundation for intensional semantics in which hyperintensional distinctions naturally emerge from the internal structure of the semantic category itself.

1 Introduction

Formal semantics has long been shaped by two major theoretical traditions. On the one hand, simple type theories (STTs), rooted in Church's λ -calculus and

developed for natural language in the *Montague* tradition, have provided a parsimonious and mathematically well-understood framework for truth-conditional meaning. On the other hand, rich(er) type theories (RTTs), following *Martin-Löf*'s intuitionistic type theory and implemented in frameworks such as type theory with records (TTR), have introduced a broader and more structure-sensitive notion of types, allowing meanings to be represented as structured objects rather than mere truth values or sets of possible worlds. Both traditions are technically sophisticated, well-established, and provide substantial empirical coverage.

Yet both traditions run into systematic difficulties when confronted with intensional and, more acutely, hyperintensional phenomena. Under intensional predicates such as *know*, *believe*, or *regret*, natural language speakers can and regularly want to distinguish expressions that the standard semantic frameworks predict to be identical, e.g., co-extensional notations, synonymous predicates, logically equivalent propositions, or active-passive pairs. The strategies that STTs and RTTs employ to accommodate such distinctions – additional input parameters in the case of STTs, more complex types in the case of RTTs – are formally coherent but laborious and often systematically unsatisfying. As argued in *Schmidt 2026*, these difficulties are not incidental failures of individual analyses but trace back to a shared foundational assumption implicit in the Curry-Howard-Lambek correspondence (CHLC) that underlies both traditions: functional extensionality. As long as meaning functions are individuated solely by their input-output behavior, intensional distinctions can only be encoded through the inputs themselves, whether through parameter proliferation or type complexity. The mismatch between an extensional formal language and a pervasively intensional object language is not a peripheral problem but a structural one.

This paper argues that embracing intensionality and giving up functional extensionality as a universal constraint, rather than patching it, offers a more principled path forward. Category theory provides exactly the formal machinery needed to do this. Originally developed as a unifying language for mathematics, category theory replaces sets and functions, both strictly extensional notions, with the more general and deliberately abstract concepts of objects and morphisms. Crucially, morphisms in a category are not automatically individuated by their input-output behavior. The identity of an object is instead determined relationally, through the totality of its connections to other objects, as captured formally by the Yoneda lemma. This makes the framework inherently suited to intensional domains, where meaning is not exhausted by extension and context is not just another input.

The present paper should be conceived as part of a larger ongoing project whose aim is to develop a fully category theoretic semantics for natural language. In

Schmidt 2026, I have aimed to work out the foundational background regarding CHLC, in particular, the formal consequences of functional extensionality and its alternatives within CHLC, as well as the notion of functional completeness and its role in structuring the semantic category. The goal of the present paper is complementary: to make the category theoretic approach accessible to a linguistically oriented audience, to motivate it against the background of familiar challenges in formal semantics, and to demonstrate through concrete analyses that the framework yields empirically adequate and structurally transparent results. The paper is thus not intended to deliver a full formalization of the system but a motivated introduction with first analyses, intended to show that the approach is both principled and applicable.

Concretely, the paper proceeds as follows. Section 2 examines how STTs and RTTs respond to intensional and hyperintensional challenges, showing that their shared reliance on functional extensionality forces structurally analogous, and ultimately unsatisfying, design choices despite their otherwise deep differences. Section 3 introduces the basic picture of the proposed category theoretic semantic framework: the distinction between an extensional and an intensional domain, the role of morphisms and compositionality, propositional structure via diagrams and functors, and the Yoneda-based notion of structural identity. Section 4 applies the framework to attitude verbs, providing analyses of factivity as a structural constraint on morphisms and of perspective-dependence as a property of source-induced intensional subcategories. Section 5 concludes and briefly points to directions for future work.

2 Type theories under intensional scope

The Curry-Howard-Lambek correspondence (CHLC) establishes the well-known relationship between logic, computation, and category theory, showing unique isomorphy between combinatory proofs, typed λ -calculus, and cartesian closed categories (CCCs), cf. Howard 1980, Lambek 1980. In natural language semantics, reflections of CHLC can be found in works on and applications of type theory, with two major influential strands: simple type theories (STTs) and rich(er) type theories (RTTs).¹ While STTs and RTTs fall into distinct domains of CHLC and follow rather different formal layouts, they both adhere to the same assumption implicit in CHLC itself: functional extensionality. In this section, I will examine how the internal logic of the two type theoretical strands answers to challenges of (hyper-)intensionality. I will show that against the background of the common

¹Cf. Sutton 2024 for a general overview.

implicit assumption of CHLC, both STTs and RTTs show similar analytic reflexes in hyperintensional settings despite their fundamental structural differences. In particular, I will claim that it is functional extensionality forcing them to resort to laborious or even inconsistent design choices in the formal semantic apparatus.

2.1 Simple type theories

Standardly, STTs assume a basic set of types and a set of rules to form complex types. Based on Church's λ -calculus and developed for natural language semantics in Montague Grammar, semantic analysis in this strand of type theory uses these types to model the meaning of natural language expressions through functions, given as typed λ -expressions, following the intuition of a correspondence between types and syntactic categories. In most analysis, the basic types include e (entities) and t (truth values), while complex types are defined inductively: If α and β are types, then $\langle \alpha, \beta \rangle$ is also a type.² Generally, $\langle \alpha, \beta \rangle$ should be understood as a function type, i.e., as a function with a domain of type α and a range of type β . Note that the constants e and t can also be described as zero-valued functions, thereby allowing to treat all types as functions equally. Standard examples of complex types are $\langle e, t \rangle$ for predicates like *(be) female* or intransitive verbs such as *laugh*, or $\langle e, \langle e, t \rangle \rangle$ for transitive verbs such as *love*. Hence, all meaning terms are functions over e and t , and compositionality of such meaning terms is modeled by function composition. Expressions are then related to a model by function evaluation (i.e., β -reduction). The setup thus falls under the Curry-domain of CHLC and exemplifies the direction "from types to propositions".

However, with equality of meaning expressions standardly being understood as co-extensionality, STTs face a first challenge by identifying meaning as the output of functions as defined above. All true (and reps. all false) and thereby co-extensional propositions collapse into a single value of type t not representative of the content of the proposition and the resp. functions that constitute it. To distinguish propositions intensionally, i.e., as representing the Fregean sense instead of solely the reference, standardly an additional intensional type s is introduced. For every extensional type α , the system thus proceeds with defining the intensional type $\langle s, \alpha \rangle$. In general, s is represented by a functional parameter λw , ranging over the set of possible worlds $\mathcal{W}$. This intensional shift renders the extension of a proposition p the set of worlds in which p is true, cf. Lewis 1986,

²While this so-called dual type system is the most widely used approach in formal semantics, it has been argued that a single type might suffice as well, cf. Partee 2009 and Liefke 2015, Liefke & Werning 2018. However, as the following considerations apply to the general logic of all STTs equally, I will limit myself to the more broadly employed dual system.

instead of its truth value alone.

Notably, it is exactly this mechanism of adding additional input parameters that constitutes the basic procedure of STTs to deal with co-extensional terms needing to be distinguished with any kind of fine(r) granularity. Crucially, the semantic system naturally still adheres to co-extensionality being the relevant notion of equality, predicting meaning terms to be equal iff they denote the same set of possible worlds. However, it is of course possible for different meaning terms or compositions of them to embody the same input-output behavior even with the additional parameter λw . For equal inputs, the functions in (1) will all have the same set of worlds as outputs.

- (1) a. $\lambda w. \text{Elton John}() = \lambda w. \text{Reginald Kenneth Dwight}()$
- b. $\lambda w \lambda x_e. \text{bachelorette}(x) = \lambda w \lambda x_e. \text{unmarried}(x) \wedge \text{woman}(x)$
- c. $\lambda w \lambda x_e \lambda y_e. \text{begin}(x, y) = \lambda w \lambda x_e \lambda y_e. \text{commence}(x, y)$
- d. $\lambda w \lambda x_e \lambda y_e \lambda z_e. \text{give}(x, y, z) = \lambda w \lambda z_e \lambda y_e \lambda x_e. \text{receive}(z, y, x)$

Apart from these notational or lexical instantiations of hyperintensionality, there are other variants of functions with necessarily co-extensional outputs, i.e., predicted to be true in exactly the same possible worlds.

- (2) a. $2 + 2 = 4$
- b. Nondeterministic space complexity classes are closed under complementation.
- (3) a. the belief that p
- b. the belief that $p \wedge (q \rightarrow p)$
- (4) a. Casey seduced the bachelor.
- b. The bachelor was seduced by Casey.

Both the analytic propositions in (2) are, being tautologies, true in all worlds, thereby denoting $\mathcal{W}$ as their extensions despite their different subject matters. A similar but contingent version is depicted in (3) with logically equivalent propositions, that differ in explanatory structure, their logical ground. And lastly, the system would consequently predict both actives and passives as in (4) to be co-extensional, as they are true in the same set of worlds.

While in extensional domains the system provides an expected and convincing analysis, it has been noted that in intensional domains, natural language is able to discern expressions that would, on the basis of their functional extensions being equal, be predicted as indistinguishable (cf. Pollard 2007, Lappin 2015, Asudeh & Giorgolo 2020, Cooper 2023). In the scope of intensional predicates, speakers can

and often need to discriminate fine-grained notions of meaning not provided by possible worlds.³

- (5) a. Mary Jane loves Peter Parker.
b. Mary Jane loves Spiderman.
- (6) a. Jim knows Phil is a groundhog.
b. Jim knows Phil is a woodchuck.
- (7) a. John believes that every prime number is divisible only by itself and one.
b. John believes that if $A \subseteq B$ and $B \subseteq B$ then $A = B$.
- (8) SCENARIO: Maddison, a well-trained logician, is certain of the proposition p and knows that p and $p \wedge (q \rightarrow p)$ are equivalent. Avery, somewhat stubborn, is also certain of p , but is skeptic about $p \wedge (q \rightarrow p)$, as it looks different to him.
 - a. Maddison is certain that p and $p \wedge (q \rightarrow p)$.
 - b. Avery is certain that p and $p \wedge (q \rightarrow p)$.
- (9) SCENARIO: Robin and Ana discuss the relationship of their friend Greg. Robin thinks that loving essentially means trusting someone. Ana thinks that loving essentially means supporting someone. Greg trusts his girlfriend, but does not support her.
 - a. Robin correctly believes that Greg loves his girlfriend.
 - b. Ana correctly believes that Greg does not love his girlfriend.
- (10) SCENARIO: Society A criticizes women initiating flirtatious contact. Society B criticizes men yielding to such flirtatious invitations.
 - a. Society A regrets that Casey seduced the bachelor, but does not regret that the bachelor was seduced by Casey.
 - b. Society B regrets that the bachelor was seduced by Casey, but does not regret that Casey seduced the bachelor.

For all (5) to (10), the embedded content is, contra intuition, predicted to be co-extensional or contradictory in the case of (9). Under intensional predicates such as *know*, *believe*, or *regret*, this is a rather unwelcome generalization, as, for instance, it might be very much plausible for Mary Jane to love Peter Parker, but not to love Spiderman, to believe some proposition p , but not $p \wedge (q \rightarrow p)$ etc.

³Examples (5), (6), and (7) resp. cited from Asudeh & Giorgolo 2020, Pollard 2007, Lappin 2015.

The hyperintensional phenomena above all share that the objects embedded under the relevant intensional predicates are co-extensional, i.e., they denote the same set of worlds. However, co-extensionality as based on equal sets of possible worlds does not exhaust the differences speakers can easily distinguish.

For STTs to represent these sometimes very fine-grained differences in meaning, they would then need to introduce new types, additional input parameters feeding into functions. For our examples, these additional parameters would need to distinguish names/guises (5), concepts (6 and 9), (explanatory) structure/ground (7 and 8), and event perspective (10). Moreover, there might be other contextual factors impacting on the meaning terms. In fact, Cooper 2005a: 344 draws attention to the possibility that there might be indefinitely many context-dependencies. While in reality the number of relevant contextual factors is finite and probably rather limited, the formal system faces several problems. First, in the presence of possibly infinitely many input parameters, the functions representing meaning are no longer well-defined. Second, the blow up in parameters dilutes the difference between regular arguments and contextual factors, as they all feed into the functions uniformly. Lastly, contextual influences might not always all be present simultaneously for a given scenario, but for reasons of well-definedness functions always have to represent them, even when parametrizing some inputs with default values. Thus, the system formally represent a worst-case scenario regarding the complexity of contexts.

It is worth highlighting that the functions themselves always adhere to extensionality. That is, functions are fully determined by their input-output behaviour: two functions are understood to be equal iff they yield the same outputs for equal inputs. To ensure functional extensionality, the number of inputs quickly proliferates when contexts call for fine-grained distinctions. Thus, it is the input of functions the contextually induced complexity is shifted to. In STTs, intensionality and the effects it produces are systematically realized through additional input parameters.

2.2 Rich(er) type theories

Rich(er) type theories, falling under the Howard-side of CHLC, exhibit a fundamentally different view of types and their function. While in STTs the set of types is, due to the inductive definition, principally infinite, only a very limited set of types building up propositions is practically used in analysis. Contrarily, in RTTs, following intuitionistic type theory of Martin-Löf 1984, propositions are considered types themselves, rather than being built up by a set of basic types. Thus, in all variations of RTTs, the set of types actually featured in semantic analysis

expectedly is considerably larger than in STTs. The variety of types is attained by either greatly expanding the set of basic types with a limited set of type constructors (cf. Ranta 1994 and subsequent work) or by using a limited set of basic types, but a large number of type constructors. Linguistics applications close to the model theoretic tradition such as TTR (type theory with records, cf. Cooper 2005a,b, Cooper & Ginzburg 2015, Cooper 2023) fall into the latter category. The record types are used to store different levels of semantic information by being structured into fields with label-type pairs such as, e.g., $[x : Ind]$, meaning the value assigned to the label x is of the type Ind (ividual). Propositions are given as (dependent) record types and made true by records (i.e., situations) witnessing the content as a truthmaker, that is, a proof.

As a consequence, meaning given by these rich types is inherently intensional due to the records being structured. Therefore, the meaning of expressions is not solely defined through their extensions in a model. In particular, two (record) types, i.e., two propositions, can then have the same witnesses, but still be distinguished via their structure. For instance, records with Phil being a groundhog or a woodchuck are different, but may share a proof (depicted by the records in 12) making the propositions true. In this case, the situation s is a proof for both propositions featuring Phil being a groundhog and Phil being a woodchuck.

- (11) a. $\left[\begin{array}{l} x = phil \\ c_1 = groundhog(x) \end{array} \right]$
 b. $\left[\begin{array}{l} x = phil \\ c_2 = woodchuck(x) \end{array} \right]$
- (12) a. $\left[\begin{array}{l} x = phil \\ c_1 = s \end{array} \right]$
 b. $\left[\begin{array}{l} x = phil \\ c_2 = s \end{array} \right]$

It is a special feature of RTTs that types are not restricted to the semantic meta language, but are allowed to feature in the object language. Thus, types can be treated as first class objects, allowing, e.g., record types (propositions) as the object of attitudes like *believe* or *know*, such as in the TTR analysis of (6a) below.

$$(6a) \quad \left[\begin{array}{l} y = jim \\ c_3 = know \left(y, \left[\begin{array}{l} x = phil \\ c_1 = groundhog(x) \end{array} \right] \right) \end{array} \right]$$

As propositions are modeled as record types, they are inherently intensional and thus not risking collapsing propositions into a single truth value. Moreover, the

structured records allow to distinguish lexical and notational varieties of hyperintensionality quite easily. Therefore, the structure in records in TTR offers an alternative to possible worlds. This immediately allows us to distinguish the (embedded) propositions in (5-7). However, for explanatory or conceptual versions of hyperintensionality as in (8-10), the analysis is less obvious.

With respect to functions, it is noteworthy that TTR is embedded in a general type theory that includes functions and function types in a version of λ -calculus with records or record types as inputs and thereby ensures compositionality following Montague's original work. Therefore, both STTs and (this version of) RTTs use λ -calculus to model meaning functions, but differ in their strategies to stratify the complexity of inputs. STTs implement the analysis of any additional complexity by adding additional input parameters, thereby changing the arity of the relevant function. As RTTs allow a significantly broader range of types, they do not add additional input parameters, but make use of more complex types, thereby circumventing the need to change the arity of the function in question. Interestingly, the necessity for context to occur in the domain of functions is one of the main reasons for the introduction of records in the first place: records can include both "the context and the propositional judgement" (Cooper 2005a: 336). Modeling hyperintensionality in TTR then can essentially be seen as a provision of the relevant additional fine-grained information communicated by a context. In fact, TTR allows content to depend on context via so-called families of record types, i.e., functions from context (records) to content (record types), for which Cooper mentions existential presuppositions as an example of these dependencies. In particular, it is worth noting that "constraints on context are indefinitely extendable" (ibid., 344). Therefore, TTR ensures well-definedness of functions modelling any number of dependencies. Additional contextual sensitivities of meaning can feed into records infinitely as they can have any number of fields.

However, there are two noticeable similarities between TTR, a representative from RTTs, and the simple typed λ -calculus from 2.1 in the Montague style, an STT. First, as records are unordered, there is no virtual difference between contextual parameters and these of classical arguments. This was also the case for function parameters in STTs. Second, just as for STTs, any additional complexity induced by hyperintensionality is shifted into the input of functions. While STTs shift the complexity towards separate additional input parameters, in RTTs it is a more complex type carrying the weight of increased fine-grainedness. This is especially noteworthy, as, just like in STTs, the main mechanism ensuring compositionality is given through functions adhering to functional extensionality. While STTs might even face the unavailability of well-defined functional possibilities

catering any number of context dependencies, the rich type system circumvents these issues by allowing types of arbitrary complexity. However, the proliferation of types both in quantity and in quality renders the notion of typehood somewhat arbitrary in a way making it difficult to generalize over structures. Both strands of type theory thus are, apart from falling under different domains of CHLC, very similar in their approach to hyperintensionality – complexity induced by intensional domains is realized by the input of extensional functions. It then follows that despite their different formal configurations, they share a common design feature lying deeper than the question of typehood itself that explains the similarity in analysis.

2.3 The hidden assumption of extensionality

Having identified the shared underlying assumption, it is worth making it once again fully explicit before considering the consequences of its relaxation. As indicated in [Schmidt 2026](#) both STTs and RTTs are established on the basis of CHLC and thus follow the main principle of functional extensionality.

(13) Functional extensionality

If $f(x) = g(x)$ for all x in the domain of functions f and g , then $f = g$.

As functions are identified solely through their input-output behaviour, they cannot directly incorporate the context they are evaluated in. If contextual dependencies are to be incorporated into meaning computation, there is no other possibility for these functionally extensional systems than to realize these contextual influences through the input of functions, may it be via additional separate input parameters, or types of increased complexity. Therefore, the detour of semantic systems realizing (hyper-)intensional meaning analysis via input modification is not incidental, but a structural feature inherited from CHLC, the more general framework both STTs and RTTs fall under.

While functional extensionality is a formal property expected and even desirable for purely extensional contexts such as calculus, programming, or purely extensional meaning contexts, it proves somewhat inept and even unnatural to assume in intensional domains of natural language. Speakers naturally situate meaning computation in contexts. As functions in the general sense cannot do that, formal systems have to resort to feeding the context as input. Since the object language itself more than often is non-extensional, an extensional formal language is bound to run into difficulties in intensional contexts, something already noted in [Peruzzi 2006](#). Thus, although it has not been noticed before, it is actually a rather straightforward observation that it is the mismatch between

an extensional formal language and an intensional object language that is fundamental. Hence, I argue that it might not be a disadvantage to allow meaning functions to be non-extensional, but that formal systems modeling natural language semantics might actually benefit from it. Therefore I propose to give up on strict functional extensionality and enrich intensional meaning terms *inherently* by their context, not just through input. Due to its inherently relational perspective, category theory emerges as the natural option to achieve this.

3 Category theory for natural language semantics: The basic picture

The discussion in section 2 has identified a structural mismatch between the formal and the object language: while natural language meaning is pervasively intensional, the formal frameworks used to model it are committed to functional extensionality. I have argued that this commitment is not a peripheral feature of individual theories, but a design choice inherited from the logical foundations both STTs and RTTs share. The question that arises is therefore not how to patch existing frameworks with additional parameters or more complex types, but whether a different formal foundation might avoid the mismatch altogether. Category theory offers exactly this. Originally developed in the 1940s as a unifying language for mathematics, it has since proven fruitful across a wide range of formal disciplines. Interestingly, although it is a more abstract perspective, it is precisely this higher level of abstraction that will allow us to be more concise.

Abstracting away The central move in category theory⁴ is to replace the notion of sets and functions operating on their elements with the more general notion of objects and morphisms between them. What makes this move significant for semantics is that objects in a category are, in a precise sense, anonymous: they have no internal structure that is directly accessible. This intentionally abstract setup establishes set-like objects and function-like morphisms without them actually being sets and functions, two concepts both strictly adhering to extensionality. An object is only ever characterized through the morphisms it participates in, i.e., the relations it stands in to other objects, c.f. Mazur 2008. Functions being strictly extensional have proven to be both too specific and too unspecific for intensional semantics. The mere identification of meaning functions through

⁴The standard textbook is Mac Lane 1998, but see also Pierce 1991 or Riehl 2016 for accessible introductions. For a short, but linguistically oriented introduction, cf. Chapter 3 in Asudeh & Giorgolo 2020.

their input-output behavior only states **that** something is true, but does not reflect **why** something is true.⁵ Functions in the world of sets are perspectively restricted to their inputs and blind towards to their contexts. However, intensional domains inherently challenge computation to be sensitive to its evaluation context. As we shall see, giving up the equality-based perspective on both inputs and functions in favor of a relational perspective on identity is not a limitation but a crucial feature that is particularly well-suited to intensional domains. Importantly, this relational setup allows us to "refine the link between intension and variational parameters" (Peruzzi 2006: 434) without giving up compositionality. This is especially worth noting since, e.g., Kratzer 2022 discusses how her set-based (and thus extensional) analysis can no longer establish compositionality: "The intensions of complex expressions are no longer always predictable from the intensions of their parts and the way they are put together." (ibid., 46). This strongly suggests that for the analysis of intensional meaning, formal systems have to give up either on compositionality or on extensionality, and I think it is a lot more convincing to drop the latter.

Objects, morphisms, and compositionality The proposed semantic category is based on a sketch from Asudeh & Giorgolo 2020: Fig. 3.3. Similar to STTs, the objects in our basic example category are simple semantic types, and morphisms should be understood as meaning terms between these types.

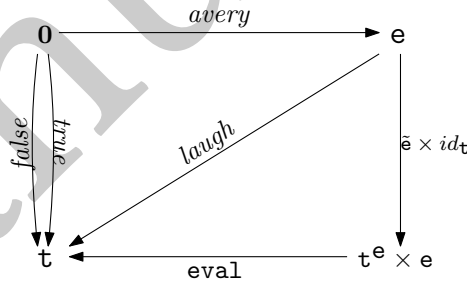


Figure 1: A basic semantic category.

As in standard STTs, the morphism *laugh* acts as a function-like map with a domain e to a range t . Note that for visual simplification, not all morphisms needed for a category to be well-defined, i.e., identity⁶ and composite⁷ morphisms, are

⁵This is also a core motivation in Truthmaker Semantics, cf. Fine 2017 for an overview. As a non-classical logic, it does not fall under CHLC and thus needs separate discussion.

⁶For every object c in a category C , there is an identity morphism $id_c : c \rightarrow c$, comparable to the identity function. Linguistically, it corresponds to the equative copula.

⁷For every two morphisms $f : a \rightarrow b, g : b \rightarrow c$, there is a composite morphism $g \circ f : a \rightarrow c$.

pictured in Fig. 1. The compositionality of meaning is modeled by the composition of morphisms: if $avery : 0 \rightarrow e$ is the morphism corresponding to the denotation of an individual, say *Avery*, and $laugh : e \rightarrow t$ is the morphism corresponding to the predicate *laugh*, then the composed morphism $laugh \circ avery : 0 \rightarrow t$ is associated with the proposition *Avery laughs*. Thus, the system does not dispose of compositionality but simply uses a different implementation of it. Predicates with a higher arity such as transitive verbs can be represented through product objects and morphisms on them, e.g., $read : e \times e \rightarrow t$. This is structurally analogous to the familiar picture in STTs, with three important differences. First, the composition is not function application in the set-theoretic sense, but morphism composition, which does not presuppose the $\in$ -based extensional perspective enforced by functions. Second, the application of predicates is not external to the category or on a meta level as in most semantic theories, but internal to the category theoretic system. This is implemented by exponential objects, representing a collection of morphisms in a category as an object in that very same category. In our case, t^e depicts the "function object", i.e., describing the function-like map with input e and output t . Together with e , we can form the product object $t^e \times e$. In a set-based world, this product would correspond to a function with domain e and range t **together** with the input e . The $eval$ morphism then corresponds to a natural way to indicate "function application" in our category, the equivalent to β -reduction of a λ -term in a set-based setup. Note these product and exponential objects must be and are indeed available in so-called Cartesian closed categories, of which our semantic category is, qua CHLC, an instance of. Third, and more importantly, two morphisms with the same source and target type are not automatically identified if they happen to correspond to the same output on all inputs. As I will make precise below, this identification of morphisms with functions, i.e., the assumption of functional extensionality in a category, is optional and not built in. In particular, extensionality and intensionality will both have their own domains, meaning that language can be constructed and interpreted locally, something that [Peruzzi 2006](#) hinted at, but did not work out. This means that the category theoretic framework can make room for an intensional domain with precisely the kind of meaning distinctions that proved elusive in section 2: distinctions between co-extensional expressions that differ in conceptual structure, inferential role, or explanatory ground.

Two domains To make this linguistically precise, I propose to distinguish two ontological domains within the semantic category. Following the considerations in [Schmidt 2026](#), where I discussed functional completeness as a composite feature consisting in representability and functional extensionality, and the signifi-

cance of that composite feature for natural language semantics, I will distinguish functional completeness in a weak and in a strong sense, each restricted to its resp. domain. In the extensional domain, the initial object $\mathbf{0}$ introduces (products and exponentials with) entities of type e and predicates on them. Entities introduced through $\mathbf{0} \rightarrow e$ are assumed to exist and thus carry existential presuppositions. In this domain, functional completeness holds in the strong sense: morphisms can be understood as functions, thus fully determined by their input-output behavior, and co-extensional meaning terms are indeed identified. This is the appropriate setting for ordinary referential expressions and truth-conditional content, where standard extensional reasoning applies without friction. On the contrary, the intensional domain i introduces intensional entities of type e^* and the predicates defined over them. In this domain, functional completeness is weakened: morphisms are still representable, but their representation is no longer unique (i.e., functional extensionality is not assumed within this subcategory). Two morphisms may realize the same extensional input-output behavior while differing, e.g., in conceptual or epistemic structure. This is not a formal defect but a deliberate design choice, reflecting the fact that intensional meaning is precisely not exhausted by extension.

The two domains are not isolated from one another, but connected in two ways. First, extensional entities of type e can be shifted into the intensional domain via a dependent factoring morphism $s : e \rightarrow i$, which allows to embed an extensional referent as a conceptual object of type e^* . Intuitively, this corresponds to moving from treating something as a referent in the world to treating it as an object of thought. Importantly, this shift is not reversible: there is no canonical morphism from e^* back to e , since not every object of thought has an extensional correlate. This asymmetry reflects the familiar observation that while every real object can be thought about, not every object of thought exists – a structural property that a possible-worlds framework, for instance, does not directly capture. Second, I assume that intensional objects are ontologically dependent on intensional predicates introducing them. In particular, expressions like *belief*, *hope*, or *knowledge* are assumed to be morphisms of type $\mathbf{0} \rightarrow i$ (cf. Fig. 2) as they each introduce their own interpretative space crucially differing from the actual reality, represented by the extensional ontological source $\mathbf{0}$. The factoring morphism s is assumed to be interpretable only against the background of the existence of a morphism of this kind to ensure that, e.g., morphisms like *phil*^{*} and *groundhog*^{*} are understood as parts of *belief* or *knowledge* or other intensional objects. However, morphisms of type $c : i \rightarrow e^*$ depicting intensional entities don't need to have a corresponding extensional variant $c^* : \mathbf{0} \rightarrow e$. Correspondingly, entities introduced (or shifted) through $i \rightarrow e^*$ intentionally do not or no longer carry

existential presuppositions. Thereby, the system also captures entities of hypothetical (*white holes*, *Planet Nine*) or impossible nature (*hexagonal circles*, *Avery's perpetuum mobile*) that cannot be shifted from existing ones introduced from 0.

Structural identity and the Yoneda perspective With functional completeness relaxed in the intensional domain, a positive account of what it means for two intensional objects to count as identical is needed. Category theory provides this through the *Yoneda lemma*, which encodes a precise and mathematically robust notion of structural identity. The key idea can be stated informally as follows:

(14) **Yoneda perspective**

An object is fully determined by the totality of its relations to all other objects in the category.

More precisely, the Yoneda lemma establishes that any object a in a category is completely characterized by the representable functor $C(a, -)$, which maps each object c to the set of all morphisms from a to c , and each morphism $h : c \rightarrow d$ to the corresponding map between hom-sets. Two objects are isomorphic – that is, structurally indistinguishable from the category's internal perspective – if and only if their representable functors are naturally isomorphic, cf. [Riehl 2016](#), Theorem 2.2.4. This means that the question of identity is not settled by checking whether two objects are literally equal, but by asking whether they stand in exactly the same pattern of relations to everything else.

For the semantics of natural language, this perspective has two important consequences. First, it replaces the binary question of co-extensional identity with a qualitative one. Identity is always identity relative to a context of relevant structural relations. Two intensional objects may be co-extensionally equivalent yet structurally distinct, if the morphisms available in the relevant intensional subcategory differentiate them. Second, and relatedly, the Yoneda perspective supports an epistemic rather than realistic notion of truth, cf. [Peruzzi 2006](#). Rather than asking whether a proposition is true in a fixed set of possible worlds, the relevant question becomes: in which relational context is the meaning computation situated, and which morphisms does that context make available? This corresponds naturally to how speakers actually evaluate intensional reports – not by checking absolute extensions, but by situating an expression within a network of inferential and epistemic relations. The relational perspective will become especially useful again when analyzing perspective sensitivity such as in, e.g., (9).

Propositional functors Lastly, something is to be said about the definition of a proposition in the system at hand. It would of course be tempting to define truth and falsity of a proposition through the way in which a diagram such as in Fig. 1 commutes. Assuming *Avery laughs* is a true sentence in the actual world, this proposition cannot simply be true by morphisms in Fig. 1 commuting. Due to commuting morphisms $true = laugh \circ avery = eval \circ \tilde{e} \times id_t \circ avery$, this would just ‘collapse’ the proposition into a single morphism. Globally, this would predict that only two propositions exist: a true one and a false one. While commuting diagrams usually seem to provide the favorite method of proof in category theory, on their own they are not useful for the analysis of natural language meaning. But indeed category theory offers an alternative preserving the structure of propositions like records in TTR do. This is achieved by strictly distinguishing between morphisms and the diagram they are in. Crucially, propositions are not identified with morphisms of the sort $0 \rightarrow t$, but with structured diagrams inducing such morphisms. In particular, morphisms with domain 0 and range t are associated with the propositions *build*, but they are not these propositions. Instead, propositions are diagrams of structured morphisms from 0 to t . Truth (resp. falsity) of these propositions then is property of these diagrams, namely that their induced composite morphism coincides with the distinguished morphism $true : 0 \rightarrow t$ (resp. $false : 0 \rightarrow t$). Crucially, multiple (in fact, infinitely many) composite morphisms can induce the same morphism $true$, but their diagrams are different: the morphisms collapse, their structural diagrams do not. Formally, diagrams and morphisms are connected through a functor, i.e., a map between categories (not objects). The category of diagrams $\mathcal{J}$ has objects such as, e.g. $0 \xrightarrow{avery} e \xrightarrow{laugh} t$. The evaluation functor $\llbracket - \rrbracket : \mathcal{J} \rightarrow \mathcal{C}$ essentially then maps diagrams in $\mathcal{J}$ on composite morphisms in $\mathcal{C}(0, t)$. Therefore, truth of a proposition structured in a diagram D is given as:

- (15) **Def.:** Let D be a diagram in $\mathcal{J}$ and $\llbracket - \rrbracket$ the evaluation functor. Then:
 D is true $\Leftrightarrow \llbracket D \rrbracket = true$

This setup can be seen as a categorial analogue of classical model-theoretic interpretation, where diagrams in $\mathcal{J}$ play the role of structured expressions and the category $\mathcal{C}$ provides their semantic evaluation as a model. This allows for a principled middle ground between STTs with the absolute notion of propositions as possible worlds and RTTs with propositions as types: only few types are needed, but structured diagrams are inherently intensional just like types in RTTs.⁸

⁸It is worth noting that despite being structured, the setup does not run into the problem of the unity of the proposition (for discussion cf., e.g., Moltmann 2020b). Composition of morphisms

4 Attitudes in a category theoretic semantics

While there is a small but growing body of literature employing category theoretic tools in linguistics, most notably through the use of monads for scope and side effects (cf. [Shan 2001](#), [Giorgolo & Asudeh 2011](#), [Asudeh & Giorgolo 2016, 2020](#), [Elliott 2023](#)), these approaches use category theory instrumentally, as a toolkit, rather than adopting it as a foundational semantic framework. However, it is worth noting that fully endorsing a category theoretic perspective does not simply provide a set of tools, but a new conceptual **framework** itself.⁹ Adopting a category theoretic perspective on semantics does not just provide new tools for existing analyses, but shifts our focus to questions of what counts as a semantic object, what identity means, and how compositionality is implemented across the two domains of $\mathbf{0}$ and $\mathbf{i}$. In what follows, I will show how this perspectival shift plays out concretely for attitude verbs, where the two-domain architecture and the Yoneda perspective developed in section 3 yield analyses of factivity and perspective-dependence that are not merely reformulations of existing accounts, but structurally different in kind.

4.1 General analysis

In accordance with the semantic category introduced in section 3, attitude verbs can be thought of as predicates taking a subject argument and an intensional object argument and yielding a truth value. More precisely, for a verb embedding an attitude such as *think*, *hope* or *know*, the corresponding morphisms are of the sort $e \times i \rightarrow t$, i.e., a domain with a product object consisting of the extensional subject entity e and of an intensional object i characterizing the attitudinal object, and a truth value t as its range. Again, as the subject argument is of the extensional type e , it is expected to be introduced via a morphism $\mathbf{0} \rightarrow e$ and thus carries the usual existential presuppositions: the attitude holder is assumed to exist. The second argument, however, is not a proposition in the classical sense. It is not a set of possible worlds, nor a truth value, nor a record type, but an element of the intensional domain i .

Precisely what kind of objects populate the intensional domain i ? Following [Moltmann](#)'s notion of attitudinal and modal objects, I take the domain i to be constituted by abstract intensional objects such as beliefs, thoughts, desires, fears, and hopes. These are not mental states in the psychological sense, but the kind of

provides a uniform mechanism to combine elements in the structure and both the composite morphism and the corresponding diagram tie them into a whole.

⁹Cf. discussion in [Król 2019](#).

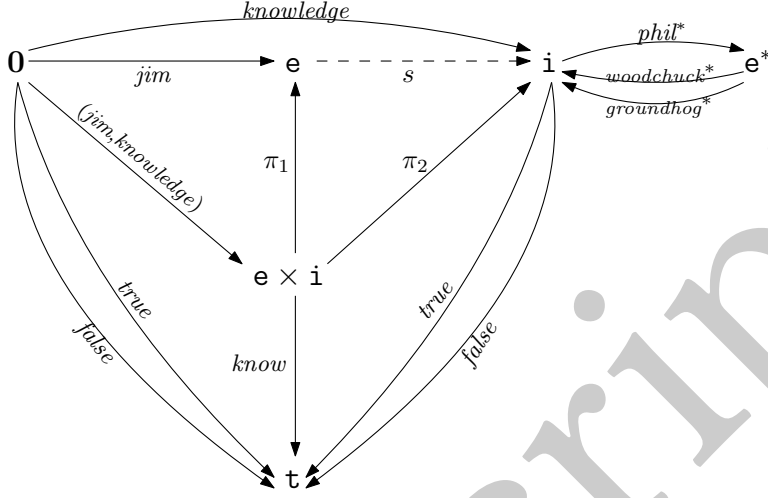


Figure 2: Semantic category with meaning terms from 6.

abstract objects that natural language attitude reports are about. In the present framework, they are introduced by morphisms of the sort $0 \rightarrow i$,^{10,11} each representing a particular type of intensional commitment: belief introduces one kind of intensional space, knowledge another, hope or desire yet different kinds of interpretative spaces. Note that the objects of intensional verbs are not e^* , but i uniformly. Thereby, the system admits both nominal (*seek a unicorn*) and structured objects (*hope that avery laughs*) in the intensional domain. The content of such a structured intensional object, i.e., a propositional unit embedded under the attitude verb, is then modeled as an endomorphism $c^* : i \rightarrow i$, characterizing the internal structure of the intensional object without reducing it to a truth value or a set of worlds.

This is a significant departure from standard approaches. In STTs and RTTs alike, the embedded proposition is evaluated and eventually typed as $\langle s, t \rangle$ (or a record type made true or false by a proof object). In the present framework, the content is embedded in i and is, intentionally, not directly truth-evaluable in the extensional sense. Truth and falsity of intensional content are represented by the

¹⁰Notably, by introduction from 0 , the attitudinal and modal objects are presupposed to actually exist. However, that does not mean that their content is supposed to be true.

¹¹Importantly, morphisms such as $0 \xrightarrow{\text{belief}}$ do not mean that *belief* is phonologically realized, as otherwise the setup would predict (pragmatically) marked phrases like *believe a belief*. Instead, these morphisms can be understood as a realization of *that* or, generally, content-embedding complementizers.

morphisms $true, false : i \rightarrow t$, which apply to the intensional object as a whole rather than to its content directly.¹² This correctly captures the intuition that it is the belief, hope, or thought that is true or false, not the embedded proposition in isolation. This analysis is based on evidence from the substitution problem, manifesting that *propositions* generally are not the kind of objects immediately embedded under attitude verbs. Speakers have beliefs, thoughts, hopes, and dreams, but not propositions.¹³

- (16) a. Maddison fears that they will not have french fries at the function.
 b. ?? Maddison fears the proposition that they will not have french fries at the function.

This is strengthened by the fact that the system requires morphisms such as, e.g., $hope : 0 \rightarrow i$ to commute with $\pi_2 \circ (avery, hope)$. Therefore, intensional objects such as *hope*, *belief*, etc. are always anchored to and dependent on their corresponding verbal predicates introducing them. This formal setup thus aims to display a constructive notion of content objects and reflects Twardowski's distinction between actions and the products created by them.^{14,15}

The following analysis of (6a) illustrates how the framework applies in practice.

- The morphisms *Jim*, *knowledge*, and the product morphism (*Jim, knowledge*) are introduced extensionally by 0 and thus presupposed to actually exist.
- The (composite) morphisms of type $0 \rightarrow i$ *knowledge* and $\pi_2 \circ (jim, knowledge)$ commute, therefore it is Jim's knowledge, created by him knowing something. Actions and the products they create are systematically tied together.
- The content of Jim's knowledge is associated with this intensional object through the endomorphism $groundhog^* \circ phil^*$, depicting the embedded proposition *Jim is a groundhog*.

¹²As pointed out in Moltmann 2017, 2020b, modal objects often are better represented by satisfaction conditions rather than truth conditions. While additional morphisms $(un)satisfied : i \rightarrow t$ are possible, they are not essential as the model-theoretic evaluation through the functor $\llbracket - \rrbracket$ can easily be adjusted to modals by replacing *true* with *satisfied* on the lefthandside of the equivalence in (15).

¹³Cf. Moltmann 2020a,b for discussion.

¹⁴Cf. Twardowski 1912 (repr. 2017) and application in, e.g., Moltmann 2017.

¹⁵Note that this does not simply reduce to a content-force distinction (cf. Hanks 2007), as this system emphasizes and requires action and product to be constructively linked. Crucially, propositional content is not assumed to exist neutrally unless actively created.

- The product object involves at least one intensional object, therefore functional completeness is weakened to representability (cf. constraint C2 in Schmidt 2026). This means:
 - The intensional entity *Phil*, introduced via $phil^* : i \rightarrow e^*$ is not presupposed to exist, and the predicate $groundhog^*$ does not automatically apply to him (not yet, cf. section 4.2).
 - In the domain i , multiple morphisms can display the same input-output behavior. Morphisms are understood as conceptual representations, not functions in the extensional sense. Therefore, $groundhog^*$, $phil^*$, and their composition are sensitive to the contexts it may be situated in.
- The proposition, i.e., the diagram for *Jim knows Phil is a groundhog*, given in 2 (without $woodchuck^*$), is true iff the following morphisms with domain 0 and range t commute:
 - $know \circ (jim, knowledge)$
 - $eval \circ (e \times \tilde{i}) \times id_t \circ (jim, knowledge)^{16}$
 - $true$

and if any additional constraints are met (cf. following sections).

4.2 Factivity

Among the most well-studied distinctions within the class of attitude verbs is the contrast between factive and non-factive predicates. Factive verbs such as *know*, *regret*, or *realize* presuppose the truth of their embedded complement: if Jim knows that Phil is a groundhog, then Phil is presupposed to actually be a groundhog. Non-factive verbs such as *think* or *believe* carry no such presupposition: in (9), Robin can believe that Greg loves his girlfriend without this committing us to any particular extension Greg actually loving his girlfriend. In standard semantic frameworks, factivity is typically treated as a lexical stipulation: factive predicates are simply defined to have this presupposition as part of their lexical entry, and the presupposition is inherited by the matrix sentence through a designated projection mechanism (cf. Kiparsky & Kiparsky 1970, Karttunen 1971, Beaver 2001 or Romoli 2015 for a pragmatic approach).

In the present framework, factivity is not a projection stipulated for individual

¹⁶These exponential objects and corresponding morphisms are, in favor of better visualization, not depicted in 2.

verbs, but rather a structural property of certain morphisms in the semantic category. Specifically, a constraint on the relationship between the intensional domain i and the extensional domain 0 . In this system, factivity induces the necessary relation of content embedded under i with the object t via a *true*-morphism and the content to be true under 0 . However, the constraint should also guarantee the entailment of embedded content regardless of the truth of the matrix predicate, thereby ensuring that both *Jim knows Phil is a groundhog* and *Jim doesn't know Phil is a groundhog* necessitate the truth of *Phil is a groundhog*. The necessary truth of the embedded content can be captured by a constraint on all factive predicates f .

(17) **F: Factivity**

f factive, $f \circ (x, y) = \text{eval} \circ (\tilde{e} \times \tilde{i}) \times id_t$ and $c^* \circ \pi_2 \circ (x, y) = c^* \circ \text{obj}_f$
 $\Rightarrow \exists c \in \text{Hom}(0, t) : c \cong c^*$ and the following diagram commutes:

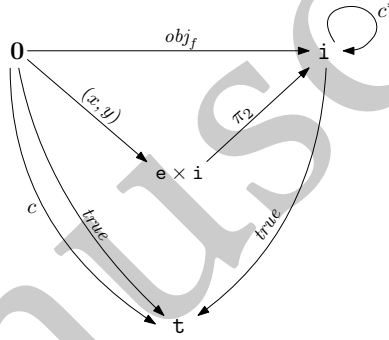


Figure 3: Commuting diagram for factivity constraint.

The antecedent consists of a conjunction with two parts. The first conjunct identifies the evaluation of a factive predicate: if $f \circ (x, y)$ commutes with $\text{eval} \circ (\tilde{e} \times \tilde{i}) \times id_t$, then the predicate f is currently evaluated on the product (x, y) . It is important to note that this constraint deliberately does neither mention commutation of these morphisms with *true* nor with *false*, i.e., it is not relevant whether f actually applies to (x, y) . The important feature is simply that it is currently evaluated. The second conjunct identifies the content embedded under f . The commutation of $c^* \circ \pi_2 \circ (x, y)$ and $c^* \circ \text{obj}_f$ ensures that it is the content associated with x 's attitudinal object obj_f . The consequence stipulates the existence of an extensional proposition c , thus given as a composite morphism with domain 0 and range t . This proposition c is extensionally associated with the content c^* associated with the intensional object i . As c^* corresponds to a conceptual representation and c to a function in the extensional sense, they cannot be compared

straightforwardly. Instead, $\cong$ is intended as a correspondence relation, associating reference and sense. $c \cong c^*$ means that the contents described through the resp. composite morphisms are equivalent modulo conceptual differences.¹⁷ Lastly, the commuting diagram itself captures the heart of the factivity constraint. If it commutes, it means that the content c is true in the extensional domain, just like the attitudinal object obj_f , associated with that isomorphical content c^* , is true. This also corresponds to *knowledge* being necessarily true, while *belief* or *hope* can be either true or false. Factivity thus emerges not from a lexical projection mechanism but from the morphism-level structure of the semantic category: the constraint captures the difference between a class of morphisms that must satisfy the commutativity condition and a class that need not.

4.3 Perspective-dependence

Beyond factivity, the category theoretic framework also offers a principled account of a subtler variety of hyperintensionality. The examples as in (9) and (10) from section 2 introduced a perspectively induced hyperintensionality, one in which the same surface expression is interpreted differently depending on the perspective of the attitude holder. Robin and Ana attribute the same predicate *love* to Greg's relationship, yet reach opposite conclusions. Society A and society B regret or do not regret the same event described by the same sentence, depending on whose perspective structures the evaluation. This type of perspective-sensitivity has been discussed under various headings in the literature, including centered worlds, *de se* attitudes, and more recently perspective-sensitive predicates analysed with monads (cf., e.g., Asudeh & Giorgolo 2016). What these cases share is that the relevant predicate does not have a fixed extension that is independent of the evaluating agent's perspective. The content is interpreted against the background of a particular source that created the attitudinal object. Interestingly, this holds both for foreign and private sources.

- (18) a. *Robin* believes that Greg *loves* his girlfriend.
 b. Robin believes *Greg* that he *loves* his girlfriend.

In (18a), it is the lexical meaning of *love* according to Robin's perspective that constitutes the content of Robin's belief. Contrarily, in (18b) we must assume that Robin believes Greg to love his girlfriend the way Greg understands the meaning of *love*. This matches the intuition that by believing someone, we take over

¹⁷One can also understand their equality by checking if c^* is one of the possible conceptual representations of c induced by a shift $s \circ c$ into the intensional domain.

their perspective. Thus, sources are crucial in evaluating meaning in fine-grained perspective-dependencies of, e.g., conceptual nature.¹⁸

In the present framework, perspective-dependence receives a natural analysis in terms of the structure of the intensional subcategory induced by a given attitudinal object. Recall that attitudinal objects such as beliefs, thoughts, or desires characterize the intensional domain i via morphisms $0 \rightarrow i$. Different agents induce, in general, different intensional subcategories: Robin's belief that loving means trusting and Ana's belief that loving means supporting correspond to different morphism profiles available within their respective intensional domains. In particular, the morphism $love^* : e^* \rightarrow i$ is not a single extensionally fixed morphism but is expected to vary conceptually under intensional scope. These variations are allowed due to functional completeness being weakened to representability without enforcing representations to be unique. Therefore, $love^*$ can simultaneously correspond to distinct criteria of application that Robin and Ana associate with *love*. For illustration, one can think of these representations as 'splitting up' $love^*$ into $love_R$ and $love_A$. Generally, for a source x creating the attitudinal object associated with content c^* , we depict c_x as the interpretation of c^* according to x 's perspective.

This is where the Yoneda perspective becomes directly relevant. The identity of an intensional object is determined not by its extension but by its full profile of relations to other objects in the category. Under Robin's subcategory, Greg's relationship stands in the relation characterized by $love_R$, because Greg trusts his girlfriend and Robin's criterion for love is trust. In Ana's induced subcategory, the same relationship does not stand in the relation characterized by $love_A$, because Greg does not support his girlfriend and Ana's criterion for love is support. The two subcategories make different morphisms available, and therefore the Yoneda profiles of the relevant intensional objects differ: they are genuinely distinct objects within their respective subcategories, even though they correspond to the same individual and the same surface predicate.

Importantly, this analysis does not treat perspective-dependence as lexical ambiguity of the predicate *love*. The predicate is not generally ambiguous, it has a uniform morphism type $love : e^* \times e^* \rightarrow i$ across all subcategories. What varies is the intensional subcategory itself, i.e., the set of morphisms available within the domain induced by a particular attitudinal object and agent. Perspective-sensitivity is therefore a structural property of the intensional subcategory, not a property of individual lexical items. Formally, this can be expressed as a constraint on the morphisms available across subcategories. Two agent-induced sub-

¹⁸Dj  r 2023 discusses the importance of sources, but does so only for distributional reasons.

categories i_a and i_b are potentially perspective-distinct with respect to a predicate f^* if the representable functors $C(f_a, -)$ and $C(f_b, -)$ are not naturally isomorphic, that is, if the Yoneda profiles of f^* differ across the two subcategories. This is not an arbitrary stipulation but a mathematically precise condition: it states exactly when two agents' perspectives on a predicate are structurally distinct in a way that generates different truth conditions for attitude reports. A more easily readable version of the constraint then can be formulated as:

(19) **P: Perspective**

Let f be a predicate with an intensional object y at position i in the product morphism τ created by x . Then:

$$c^* \circ \text{obj}_f = c^* \circ \pi_i \circ \tau \Rightarrow c^* = c_x$$

The analysis thus captures a general pattern: whenever co-extensional expressions or co-referential individuals appear in attitude contexts and generate diverging truth conditions, the explanation lies not in the ambiguity of the relevant expressions or in additional input parameters, but in the structural differences between the intensional subcategories involved. Different attitudinal objects and different agents induce different Yoneda profiles, and it is these profiles – the totality of relations an object stands in within its domain – that determine meaning in the intensional setting.

5 Conclusion

This paper has argued for a category theoretic foundation for natural language semantics, motivated by a structural diagnosis of existing approaches. Both STTs and RTTs, despite their considerable differences, inherit the assumption of functional extensionality from the Curry-Howard-Lambek correspondence and are therefore forced to encode intensional and hyperintensional distinctions through the inputs of meaning functions, via parameter proliferation or type complexity. This shared design feature is not incidental but reflects a fundamental mismatch between an extensional formal language and an intensional object language. Category theory resolves this mismatch at the level of foundations rather than by local repair. By replacing set theoretic functions with morphisms that are not automatically individuated by their input-output behavior, the framework allows intensionality to be a structural feature of the semantic system rather than a special case to be accommodated. The proposed two-domain architecture, i.e., distinguishing an extensional domain, where functional completeness holds in the strong sense, from an intensional domain, where it is weakened to representability without uniqueness, provides a principled basis for maintaining standard

truth-conditional reasoning in extensional contexts while capturing fine-grained meaning distinctions in intensional ones. The Yoneda perspective grounds a relational notion of identity suited to the latter. Applied to attitude verbs, this yields analyses of factivity as a morphism-level constraint and of perspective-dependence as a structural property of source-induced intensional subcategories, both without ad hoc stipulations.

Several directions remain for future work. On the empirical side, the framework needs extension towards *de re/de dicto* distinctions, entailment behavior of content nouns embedded under attitudes, modal verbs, and context-sensitive predicates more broadly. On the formal side, a more complete axiomatization of the semantic category is needed, as are precise conditions on the interaction between the two domains. A particularly promising next step is the move to enriched categories, which would allow morphisms themselves to carry intensional structure – a natural development of the relational perspective pursued here.

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